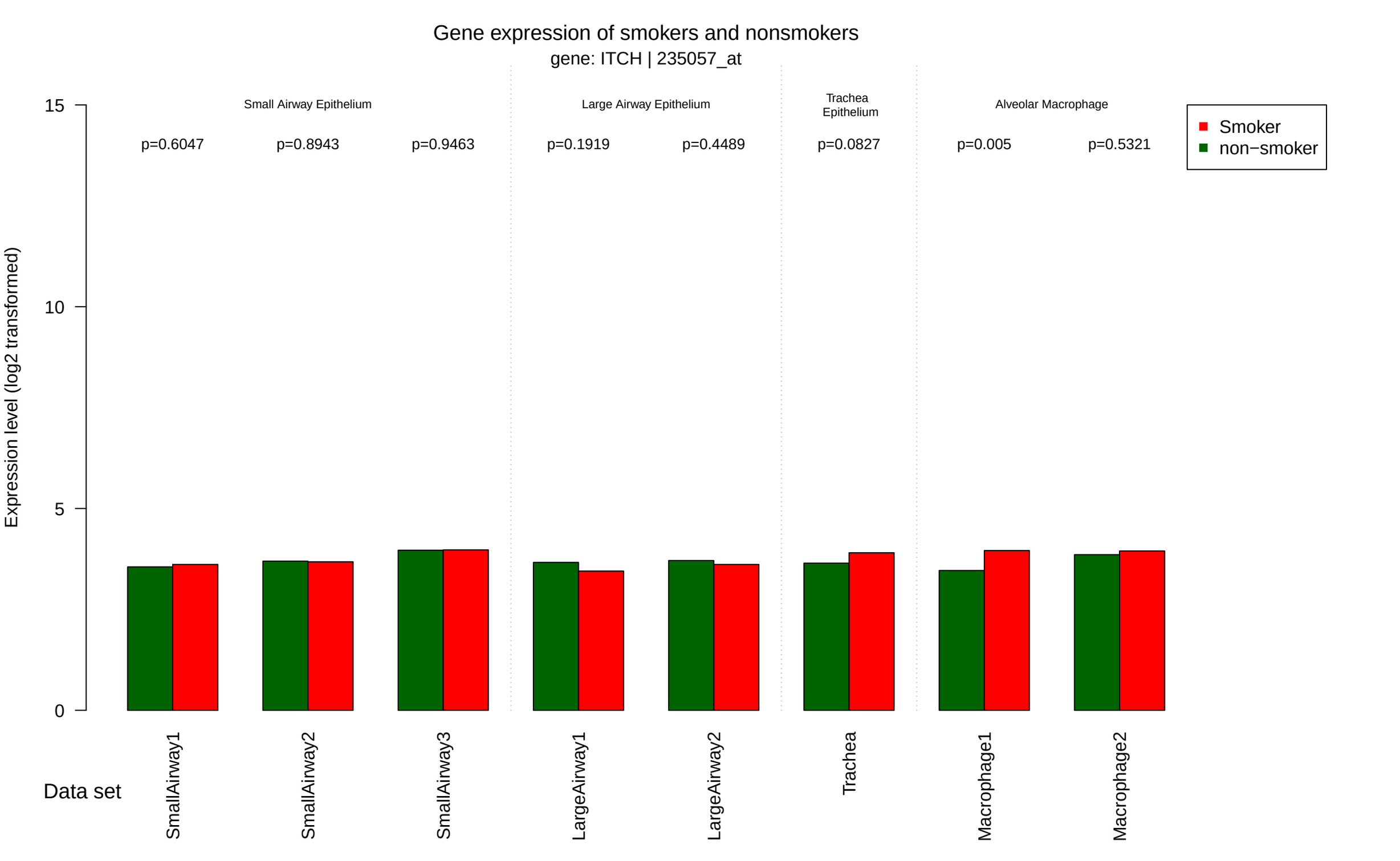
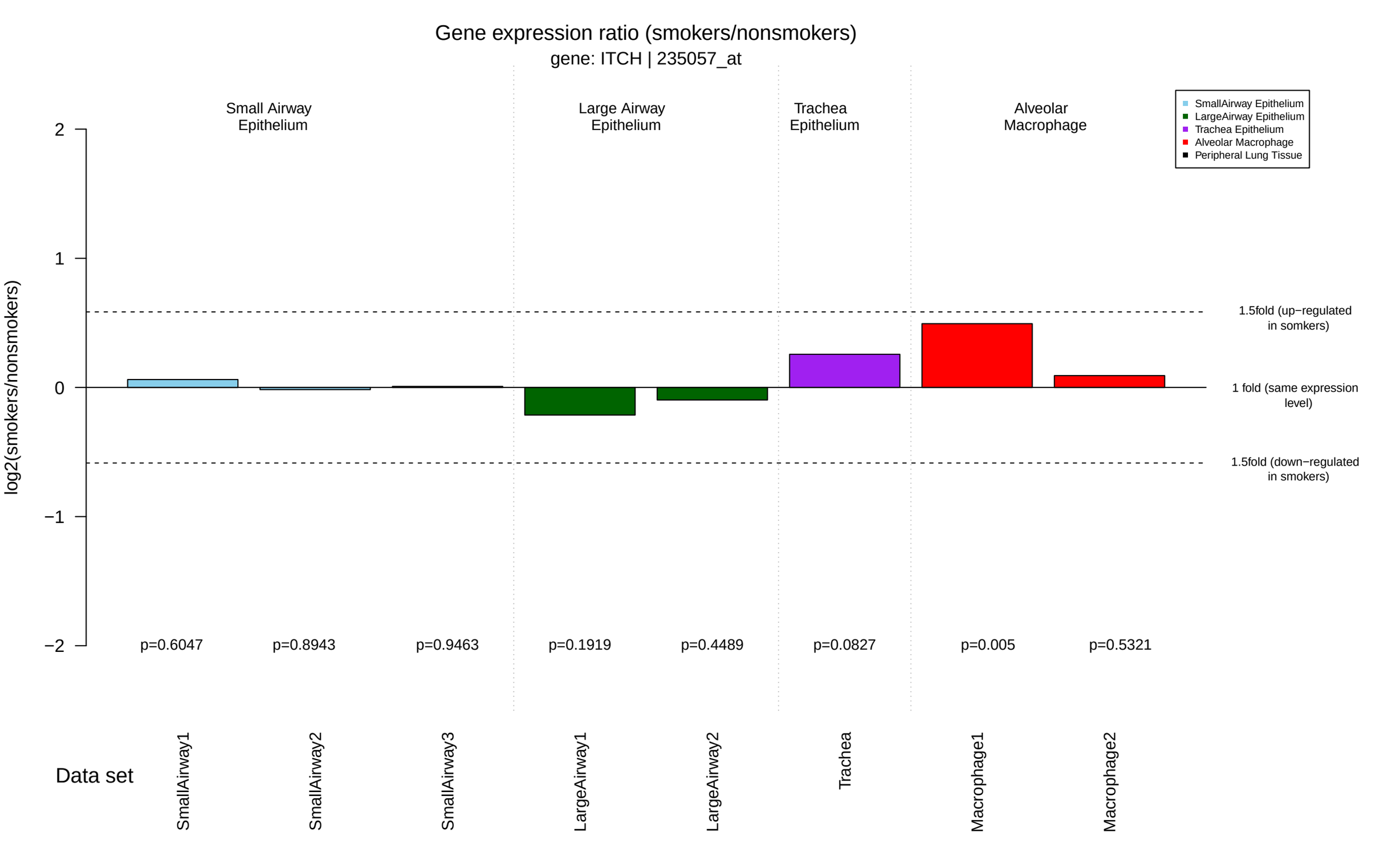
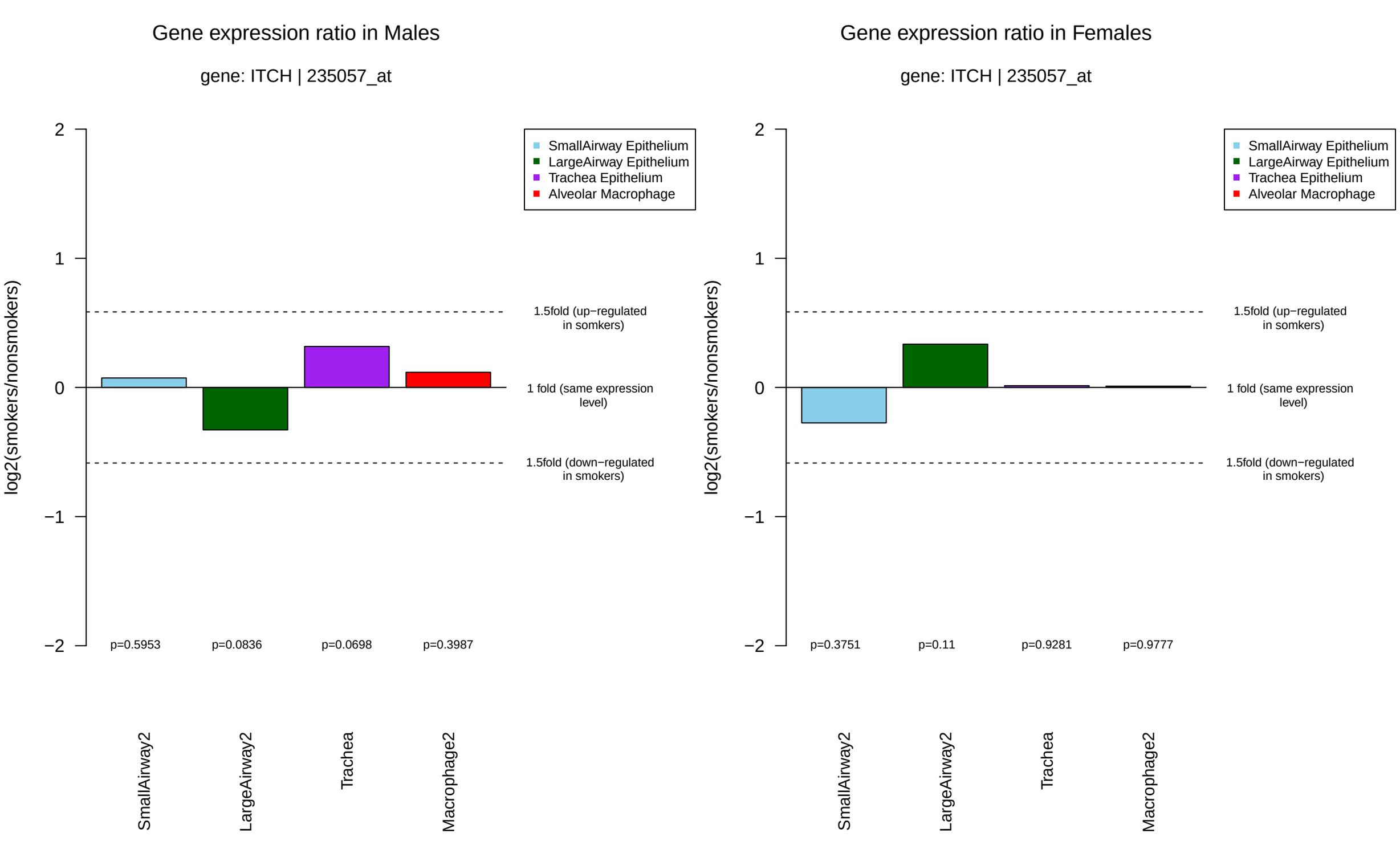
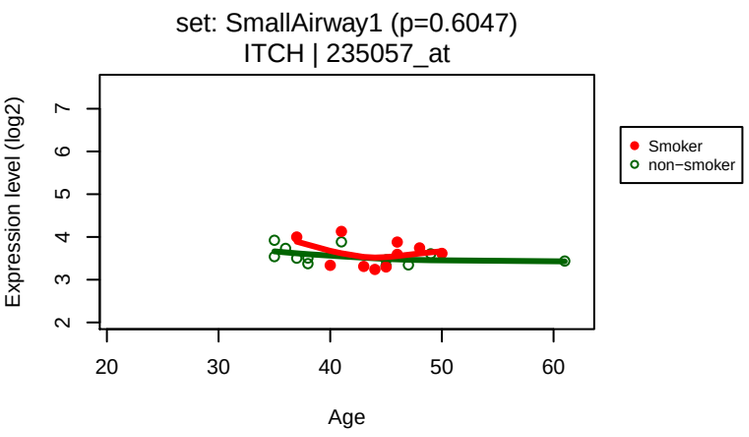
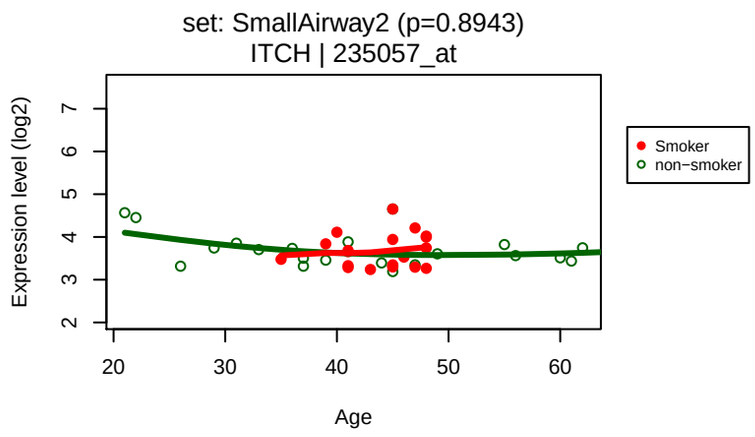
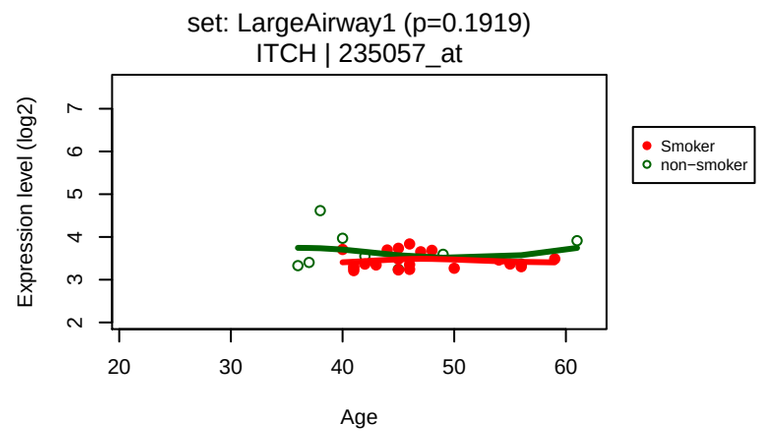

Gene expression of smokers and nonsmokers
gene: ITCH | 235057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6047
p=0.8943
p=0.9463
p=0.1919
p=0.4489
p=0.0827
p=0.005
p=0.5321
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

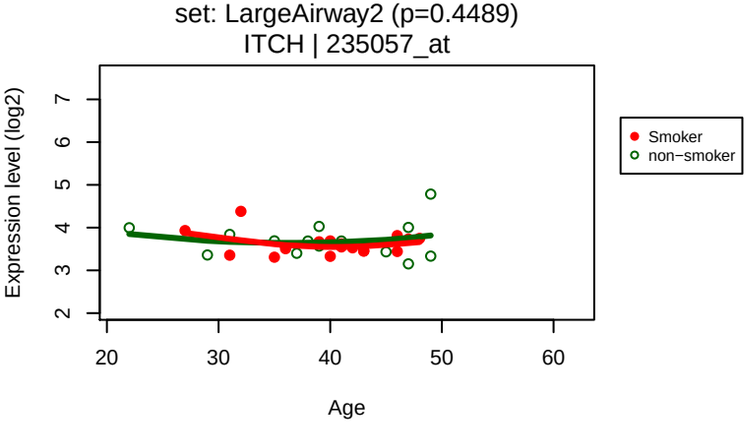

Gene expression ratio (smokers/nonsmokers)
gene: ITCH | 235057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6047
p=0.8943
p=0.9463
p=0.1919
p=0.4489
p=0.0827
p=0.005
p=0.5321
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

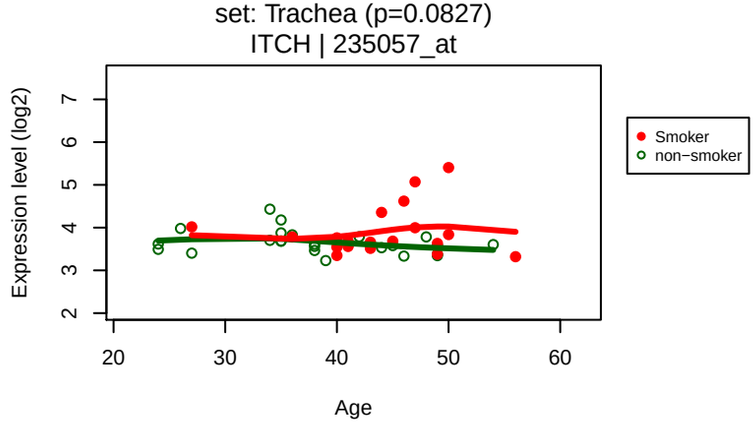

Gene expression ratio in Males
gene: ITCH | 235057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5953
p=0.0836
p=0.0698
p=0.3987
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITCH | 235057_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3751
p=0.11
p=0.9281
p=0.9777
SmallAirway2
LargeAirway2
Trachea
Macrophage2

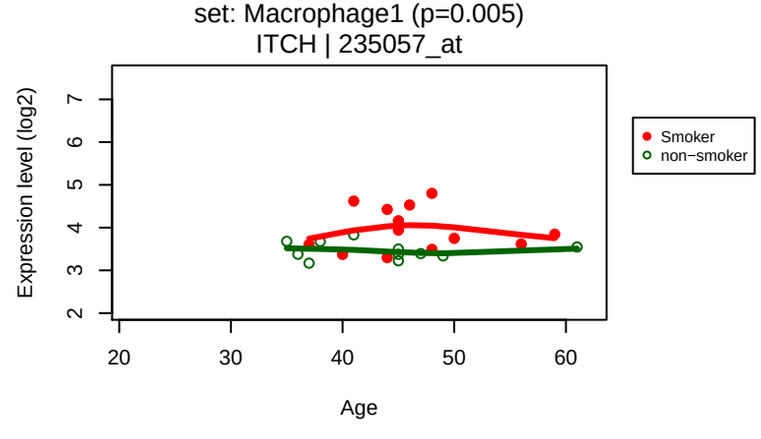

set: SmallAirway1 (p=0.6047)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker

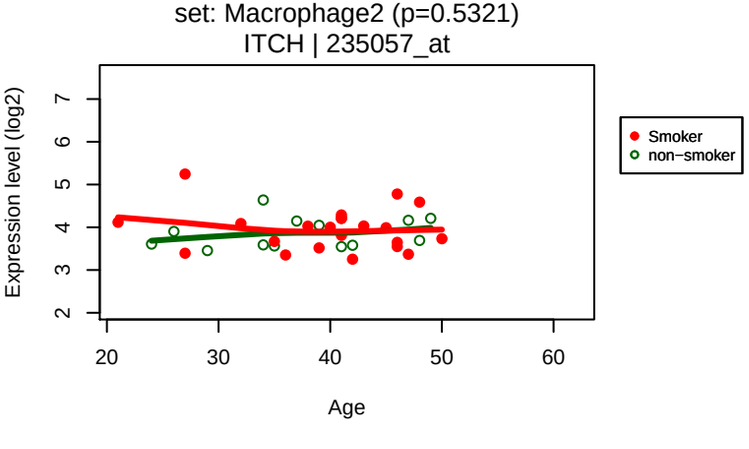

set: SmallAirway2 (p=0.8943)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker

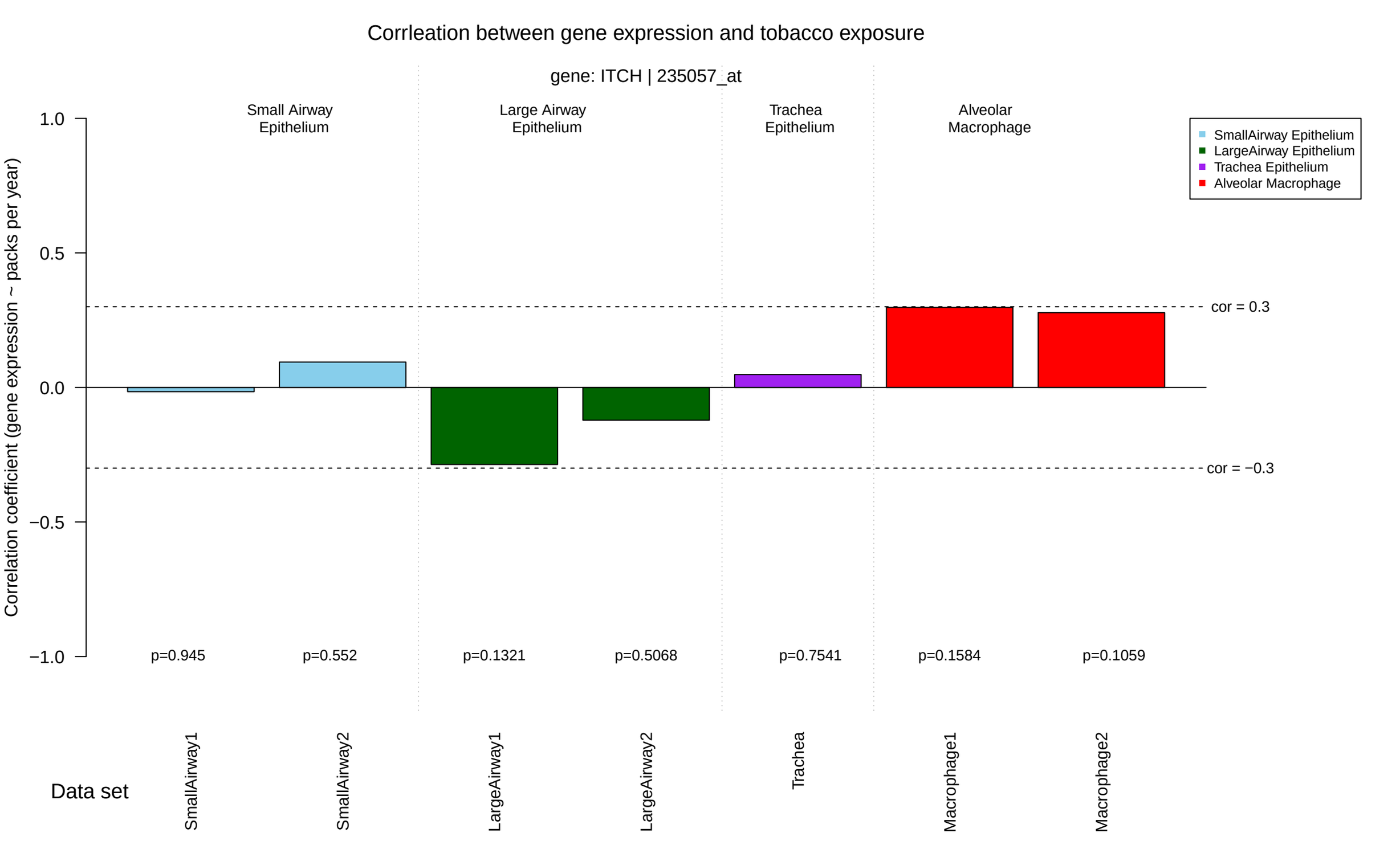

set: LargeAirway1 (p=0.1919)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4489)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0827)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.005)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5321)
ITCH | 235057_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITCH | 235057_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.945
p=0.552
p=0.1321
p=0.5068
p=0.7541
p=0.1584
p=0.1059
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2